

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

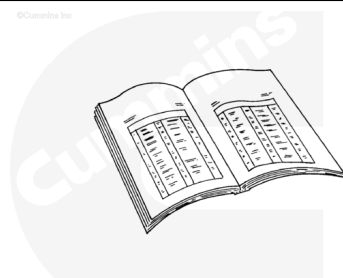
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations

⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.



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- Disconnect the batteries. Refer to Procedure 013-009 in Section (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Open the draincock on the oil cooler housing and on the water pump. Close the draincock after the coolant is drained.
- Remove the turbocharger oil drain line. Refer to Procedure 010-045 in Section (</qs3/pubsys2/xml/en/procedures/20/20-010-045.html>)
- Remove the turbocharger coolant supply hose. Refer to Procedure 010-037 in Section (</qs3/pubsys2/xml/en/procedures/20/20-010-037-tr.html>)

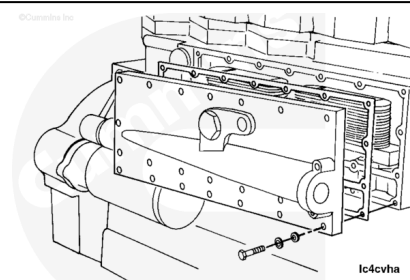
- Remove the water bypass tube. Refer to Procedure 008-062 in Section (/qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html)
- Disconnect the aftercooler supply hose from the water pump outlet connection. Refer to Procedure 010-002 in Section (/qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html)
- Remove the water pump outlet connection assembly. Refer to Procedure 008-062 in Section (/qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html)

Remove

Remove the oil cooler cover (if equipped)

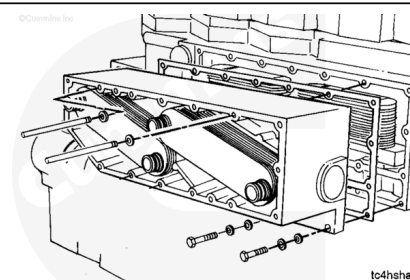
Remove all of the capscrews, the oil cooler cover, and gasket.

Discard the gasket.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift component or assembly.



Remove the torque converter cooler housing (if equipped)

Use two [3/8-16 x 12 in] guide studs to support the housing while the capscrews are being removed.

Remove the remaining capscrews, the torque converter cooler housing, and the gasket.

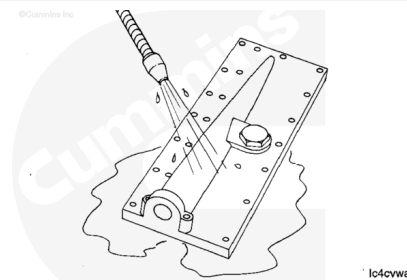
Discard the gasket.

Remove the torque converter cooler elements. Refer to Procedure 008-065 in Section (/qs3/pubsys2/xml/en/procedures/20/20-008-065-tr.html)

Remove and discard the o-rings.

Clean and Inspect for Reuse

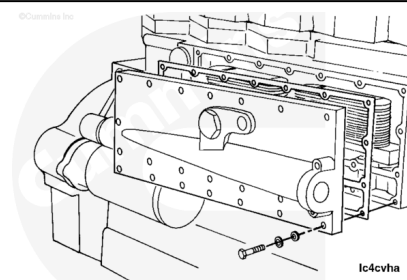
Clean and inspect the cover for reuse. Check the cast iron covers for corrosion.



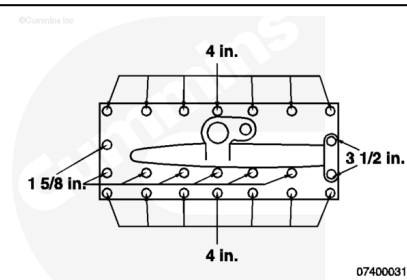
Install

Install the converter cooler housing and elements (if equipped). Refer to Procedure 008-065 in Section (/qs3/pubsys2/xml/en/procedures/20/20-008-065-tr.html)

Install the oil cooler cover and new gasket. Use several capscrews to secure the cover in position, but do **not** tighten the capscrews.

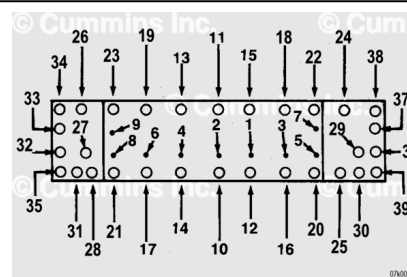


Install the capscrews as shown.



Tighten the capscrews in the sequence shown.

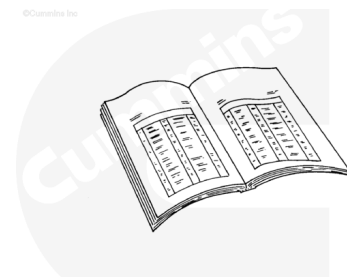
Torque Value: 60 n•m [45 ft-lb]



Finishing Steps

⚠ CAUTION ⚠

These parts are a safety-related item. The parts must be clamped correctly to prevent fretting.



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- Install the transfer tube in the water pump outlet connection. Refer to Procedure 008-062 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html>)
- Install the water pump outlet connection, gasket, and capscrews. Refer to Procedure 008-062 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html>)
- Connect the aftercooler supply hose to the water pump outlet connection. Refer to Procedure 010-002 in Section (</qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html>)
- Lubricate the o-ring on the bypass tube. Install the coolant bypass tube. Refer to Procedure 008-062 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html>)
- Install the oil drain tube and gasket to the turbocharger. Refer to Procedure 010-045 in Section (</qs3/pubsys2/xml/en/procedures/20/20-010-045.html>)
- Install the turbocharger coolant supply hose and tighten to the turbocharger. Refer to Procedure 010-037 in Section (</qs3/pubsys2/xml/en/procedures/20/20-010-037-tr.html>)
- Fill the cooling system. Refer to Procedure Refer to Procedure 008-018 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Connect the batteries. Refer to Procedure 013-009 in Section (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Operate the engine to normal operating temperature and check for leaks.

